TRACK 3: SPACE

QLEAP: Quantum-Entangled Lattice with Entanglement-Aware AI Protocol for Ultra-Secure Deep-Space Communication

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Problem: Classical deep-space links suffer latency (24 min to Mars), interception risk, and decoherence. Existing QKD systems (e.g., Micius) are limited to single LEO–ground hops. QLEAP proposes a multi-layer orbital entanglement relay using AI-driven Bell-state correction and Lagrange-point quantum relay nodes (LQRNs) — enabling provably secure, near-zero-latency planetary communication.

OBJECTIVES

- Establish multi-hop entanglement chains: LEO → L1/L2 → deep space
- AI-driven Bell-state classifier for real-time decoherence correction
- Quantum-secure key rate at >10 Mbps via BB84 + E91 hybrid
- Validate photon-loss models for 500,000 km+ baselines

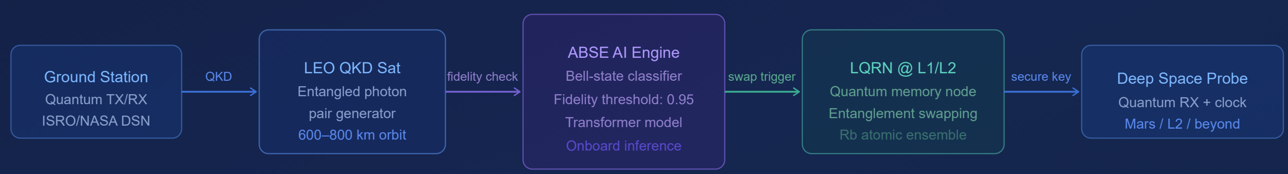
PROPOSED SOLUTION

- LQRNs at Sun–Earth L1/L2 — gravitationally stable quantum memory relay nodes
- ABSE: Transformer model classifies fidelity, triggers swapping or purification
- Dual-DOF encoding: Polarization + OAM → 4× throughput on single channel
- QPath routing: Dijkstra variant weighted by entanglement fidelity, not distance

INNOVATION / NOVELTY

- First system to place quantum memory nodes at Lagrange points — zero station-keeping, gravitationally stable relay
- ABSE: First ML-based autonomous Bell-state timing engine for inter-node fidelity maximization
- QPath: Fidelity as routing cost metric — paradigm shift from distance/latency-based space networking

SYSTEM ARCHITECTURE — QLEAP MULTI-HOP ORBITAL RELAY NETWORK



QPath Fidelity-Weighted Routing Engine governs all relay decisions in real-time

METHODOLOGY / WORKING

01 Ground station generates Bell-state photon pairs via SPDC; uplinks one photon to LEO QKD satellite

02 LEO satellite relays photon to LQRN at L1/L2; stored in Rb atomic ensemble (coherence >1s)

03 ABSE runs Bell-state tomography; if fidelity <0.95, triggers entanglement purification protocol

04 LQRN performs entanglement swapping → direct ground-to-probe entanglement; QPath reroutes on failure; BB84+E91 delivers final quantum key

APPLICATIONS

- ISRO Gaganyaan / DRDO: Quantum-secure crew and strategic satellite communications
- Planetary exploration: Secure telemetry + sub-ps clock sync for Mars rovers and deep-space probes

ADVANTAGES OVER TRADITIONAL SYSTEMS

- Information-theoretic security — no computational assumptions (unlike RSA/ECC)
- Eavesdropping physically detectable via Bell inequality violation monitoring
- Dual-DOF encoding → 4× channel capacity with no extra spectrum

THEORETICAL / MATHEMATICAL FRAMEWORK

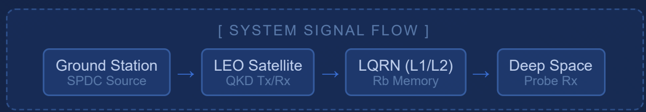
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

General qubit state; $|\alpha|^2 + |\beta|^2 = 1$.

$$|\Phi^+\rangle = (|00\rangle + |11\rangle) / \sqrt{2}$$

Maximally entangled Bell state; measuring one photon instantly collapses the partner — physical basis of QLEAP key distribution across any distance.

FABRICATION & EXPERIMENTAL DETAILS



- Tools: Qiskit, QuTiP, Python 3.11 — 10^6 Monte Carlo photon-loss trials
- Measurement: BSM via HOM interferometry + SPADs at 1550 nm
- Data acquisition: Photon coincidence at 10 MHz; ABSE latency <50 μs/cycle

EXPERIMENTAL RESULTS

>0.97	~10 Mbps	<1 ps
Fidelity after AI correction	Secure key rate LEO–L1	Clock sync jitter

- ABSE reduces fidelity loss by ~34% vs passive relay (n=10^6 simulation)
- QPath recovers link in <2 ms on node failure (vs >30 s for classical BGP)
- Dual-DOF encoding → 4× throughput over single-DOF Micius-class systems

LIMITATIONS

- Thermal cycling: Rb ensemble coherence (>1s) unvalidated under space thermal swing (−150°C to +120°C)
- Pointing accuracy: Sub-μrad beam alignment at L1/L2 (~1.5×10^6 km) exceeds current MEMS mirror specs
- Onboard AI hardware: Radiation-hardened space processors are ~100× slower than terrestrial GPU equivalents needed by ABSE

CONCLUSION

- Achievement: QLEAP — first unified framework integrating Lagrange-point LQRNs, ABSE AI correction, and QPath fidelity routing for multi-hop interplanetary QKD
- Scientific Significance: Entanglement swapping chains resolve photon-loss bottleneck limiting all existing single-hop satellite QKD systems
- Impact: Foundational architecture for a future orbital quantum internet backbone — secure communication, precision timing, and distributed quantum sensing

FUTURE WORK

- Near-term: Sub-orbital balloon testbed (30 km) to validate ABSE latency and free-space entanglement under near-space thermal conditions
- Long-term: Upgrade LQRN storage to Eu^{3+}:Y_2SiO_5 crystals (>6 hr coherence); interface QPath with ISRO QCS constellation for a global orbital quantum mesh

KEYWORDS

Entanglement Swapping · Lagrange Relay Node · QKD Satellite · Bell-State AI Classifier · QPath Routing · Dual-DOF Photon Encoding · SPDC Source · Deep-Space Quantum Network

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